



PROJECT 5 · MODULE 5 — AI STRATEGY & BUSINESS IMPACT

Use Case Discovery

AI-Assisted Back-Office for Luxury Hospitality



*Take the grind, not the craft — freeing scarce human energy
and scarce euros for what truly justifies a luxury rate.*



SECTOR
Luxury Hospitality · 5★

COMPANY SIZE
SME · 80–250

PREPARED BY
Thibault

I Strategic Framing

Cleo runs a luxury hotel and is not anti-AI — she is anti-hype. The honest answer to “*should we invest in AI?*” begins where this segment actually hurts: margins are being squeezed from both sides, costs rising faster than revenue, while what keeps the experience standing is the passion of the people who deliver it.

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The right first AI investment does not touch the guest or replace staff. It removes invisible back-office drudgery — so scarce human energy, and scarce euros, are redirected to what justifies a luxury rate.

DESIGN PRINCIPLE — AI ASSISTS, HUMANS DECIDE

II Sector & Company Size

SECTOR

Luxury hospitality
 independent 5★

The segment that feels the cost squeeze most acutely and carries the least back-office buffer.

COMPANY SIZE

SME
 ~80–250 employees

Large chains centralise finance in shared-service centres; independents do not — the burden lands on a lean team, making the pain and the ROI concentrated and visible.

This mirrors a realistic client profile rather than a generic enterprise, which keeps the business case concrete and testable.

III Stakeholders & Pain Points

STAKEHOLDER	WHAT THEY NEED	PAIN POINT TODAY	CONSTRAINT
CEO / Owner (Cleo)	Protect margin without harming the guest experience	Every cost lever risks cutting into service quality	Won't "burn budget on hype"; needs evidence
CFO / Finance Dir.	Accurate, timely month-end close	Close is slow; the team firefights instead of analysing	Limited headcount; can't simply hire
Accountant / AP	Invoices booked correctly and on time	Manual keying of supplier invoices; repetitive, error-prone	Volume spikes overwhelm a small team
Cost controller	Match cost invoices to stock	Reconciling invoices against stock is manual and lagging	Needs accuracy before it can be trusted
General Manager	Team focused on guests, not paperwork	Skilled staff pulled into admin	Service standards non-negotiable
Front-line staff	Meaningful work, not burnout	Admin overload feeds fatigue and turnover	Attrition is already structural

The same lean back-office team is both the bottleneck and the flight risk. Relieving it addresses cost and retention at once.

IV Why This Matters Now

EVIDENCE CAVEAT — READ FIRST

Most hard benchmarks below come from US / global sources, while the client context is Luxembourg / EU. They are treated as *directional*, not as Luxembourg actuals. Securing EU-specific and client-specific figures is a documented task for the market-research stage and the pilot — exactly the kind of assumption Cleo should validate before investing.

The margin squeeze is structural, not cyclical

- ◆ Across property types, gross operating profit margins declined through 2025; the post-pandemic expense base has reset higher while RevPAR growth trails inflation in most markets (*HVS, 2025*).
- ◆ Pressure is explicitly felt in luxury operations, where revenues have recovered but expenses rise faster (*Hospitality Investor, 2026*).
- ◆ Even ultra-luxury, with a ~7.5% TRevPAR rise, saw payroll per available room climb ~8.3% — costs outrunning the revenue gain (*HOTELSMag, 2025*).
- ◆ Labour is the largest, fastest-growing cost line; operators pay ~22% more than 2019 for ~7% fewer hours worked (*CRE Daily, 2025*).

The sector runs on people who are hard to keep

- ◆ Hospitality has the highest staff turnover of any major sector — commonly cited around 70–80% annually (*Hybrid Payroll; OysterLink, 2025–26*).
- ◆ Roughly two-thirds of hotels report staffing shortages (*AHLA via Escoffier, 2025*).
- ◆ Replacing one employee is estimated at ~€4,400–4,700+ in recruiting and onboarding (*OysterLink / SHRM, 2025–26*).

The back-office “tax” is large, manual, and quietly expensive

- ◆ Manual invoice processing commonly costs ~€11–14; best-in-class automation ~€2–5 — up to ~80% lower (*HighRadius; Quadient; Nanonets, 2025*).
- ◆ Average manual cycle time is ~14.6 days, compressible to ~3–5 days (*DocuClipper; Ascend, 2025*).
- ◆ ~68% of teams still key invoices manually into the ERP; under a third are automated (*DocuClipper, 2025*).
- ◆ Per-invoice handling can drop from ~15 minutes to ~3 minutes once automated (*Artsyl, 2025*).



V Candidate Use Cases

	CANDIDATE	CORE VALUE	MAIN DIFFICULTY	VERDICT
A	Invoice → accounting entry (G&A spend), human-validated	Direct labour & error reduction on a known cost pool; clean human-in-the-loop	Auto-coding to the right GL account must be reliable	SELECTED
B	OCR cost-control invoices → P&L + <i>stock movements</i>	Extends value into inventory accuracy	Matching invoice lines to internal SKU & unit conversion is genuinely hard	Roadmap · phase 2
C	Automated monthly P&L reporting + variance commentary	Visible management value, fast to show	Less “AI core”, more BI; weak as a standalone case	Roadmap · support

B and C are deferred to the roadmap, not dropped — they form a credible expansion path once A proves the model.

VI The Selected Use Case

AI-assisted invoice processing and accounting entry, with mandatory human validation.

The system captures incoming supplier invoices, extracts the structured data, proposes a draft accounting entry — account, VAT treatment, analytical allocation — and routes it to a finance team member who reviews, corrects and approves before anything is posted.

- 1 **Highest, most defensible ROI.** It targets a cost pool with well-established benchmarks, so the case rests on industry data, not optimism.
- 2 **Clear, contained pain.** The lean finance team feels it daily; relief is immediately tangible.
- 3 **Lowest regulatory risk.** Back-office accounting automation is not a high-risk category under the EU AI Act, and the human-in-the-loop design reinforces a low-risk posture.
- 4 **Protects margin and people at once.** Money saved *and* burnout reduced — aligned to the strategic thesis.
- 5 **Honest about AI's role.** A human stays accountable for every posted entry.
- 6 **Sets up the capstone.** This is the natural POC to carry into the final project.

VII Why Now, Not “Wait”

SIGNAL	READING
Cost curves have crossed	Manual vs automated per-invoice economics now differ by ~70–80% — the saving is no longer marginal.
Technology has matured	LLM-based document processing handles varied invoice layouts far better than legacy template OCR.
Adoption still low in SMEs	With under a third automated, this is early-but-proven — past the bleeding edge, ahead of the laggards.
Regulatory picture is clear	For this low-risk use case, the EU AI Act imposes no high-risk obligations, removing a common “let’s wait” objection.

Recommendation direction: not “invest at scale”, not “wait”, but run a scoped pilot — the lowest-regret way to turn directional benchmarks into client-specific evidence.

VIII Preliminary Success Criteria

To be finalised with the client. The system “works” if it can show:

- 1 **Touchless rate** — % of invoices proposed correctly with no human correction.
- 2 **Handling time per invoice** — minutes from receipt to approved entry vs the manual baseline.
- 3 **Finance hours returned per month** — time freed for higher-value work or overtime avoided.
- 4 **Accuracy / exception rate** — error rate on posted entries vs the manual baseline.

IX Assumptions to Validate

ASSUMPTION	WHY IT MATTERS	HOW TO VALIDATE
US/global benchmarks are directionally valid for Luxembourg	The whole ROI case leans on them	Source EU/LU figures; confirm with client
Invoice volume is high enough for ROI	Per-invoice savings only matter at volume	Get the actual monthly invoice count
A lean team currently keys invoices manually	The pain premise	Confirm current AP workflow with client
Auto-coding to the correct GL account is achievable at usable accuracy	Core feasibility of the value	Test in the final-project POC on sample data
Luxury turnover dynamics match segment averages	The retention half of the argument	Cross-check with EU hospitality sources

X Explicitly Out of Scope

To keep the case honest and the risk low, this use case does **not**:

- ◆ touch payroll or any HR / worker-management decision — that would change the risk profile;
- ◆ face the guest or alter the guest experience;
- ◆ post any accounting entry autonomously — a human approves every entry;
- ◆ handle stock movements or cost-control invoices in this phase — that is the deferred roadmap.

Sources

All figures are directional industry benchmarks unless stated; EU / Luxembourg-specific sourcing is a follow-up task for the market-research stage.

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